

Purpose of the Use of public reporting in Health Care

Overview

We began by identifying the focus of each reviewed article, analyzing its title, abstract, background, research question, objective, purpose, and results. We then categorized antecedents based on their purpose, which resulted in three main themes: individual-related factors, organization-related factors, and public reporting-related factors. For individual-related factors, we included aspects such as individual behaviors and health disparities. For organization-related factors, we focused on organizational behaviors and attitudes. For public reporting-related factors, we examined quality indicators, performance indicators, data utilization, and system usage. Ultimately, we organized the studies into eight themes, which will be discussed in the following section.

Individual-Related Factors

Individual Choice

Studies have explored attitudes and preferences related to public reporting data among patients in community living centers (Davila et al., 2024). Research has also investigated the impact of public reporting on healthcare quality, particularly examining the potential positive and negative effects of report cards on the behaviors of doctors, hospitals, and patients (Werner & Asch, 2005). Additionally, some studies have focused on differences in public reporting preferences between HRC users and non-users (Emmert & Wiener, 2017). Other studies aim to understand how public reporting influences individuals' healthcare choices; for example, social media review systems (Lagu & Greaves, 2015) have significantly enhanced individuals' understanding of healthcare performance. Systems such as the nursing home 5-star rating system (Werner et al., 2016), quality ratings for home care institutions (Jung et al., 2016), and physician rating websites (Emmert et al., 2013) have all been examined for their impact on personal healthcare decision-making.

Health Literacy

Additionally, studies have analyzed how different cognitive backgrounds (Emmert et al., 2013) and decision-making contexts (Shaller et al., 2014) affect consumer understanding and decision-making related to public reporting. Factors such as data consistency, report format (text or chart), and the use of confidence intervals have been examined for their influence on consumer comprehension and decision-making (Mazor KM et al., 2009). Research also examined the impact of various

presentation formats, such as symbols and charts, to identify methods that improve user comprehension (Sander et al., 2015). Studies have considered the impact of quality reports on the care of patients with advanced dementia (Konetzka et al., 2015), and factors affecting elderly chronic patients' perceptions of physician quality (J. Christianson et al., 2014). Community studies have explored ways to use public reporting to reduce health disparities and improve awareness among minority groups (Yonek et al., 2014).

Organization-Related Factors

Attitudes

Studies on organizational attitudes have often included physicians, as their behaviors and attitudes can represent those of the healthcare institution they work for. Most research has focused on policies and data within public reporting systems. Studies have examined the perspectives of healthcare system stakeholders on the quality policies related to public reporting (Van Wilder et al., 2021), as well as the attitudes of PICU directors towards benchmarking, public reporting, and pay-for-performance policies (Natale et al., 2011). Physician attitudes toward patient experience data made public online (Lagu et al., 2019), the perspectives of European infection control stakeholders on reporting HAI data (Martin M et al., 2013), and the attitudes of CLC caregivers toward public reporting systems (Lloyd et al., 2021) have also been analyzed.

The phenomenon of “metric-driven harm” has been analyzed through case studies to examine its effects on healthcare outcomes (Rambur et al., 2013), while the financial costs and administrative burdens of PCI result reporting programs on hospitals and physicians have been investigated (Wadhera et al., 2019).

Policy and Behavior

This section explores how transparency within collaborative projects contributes to quality improvement and reduced mortality rates. Studies have offered practical recommendations for enhancing quality competition in Germany (Negele et al., 2023) and examined the impacts of public reporting transparency on patient choice, hospital investment, social welfare, and healthcare market structures (Saghafian & Hopp, 2020) (Lihn et al., 2015). The studies further investigate the

foundational assumptions of mandatory public reporting and assess its potential effectiveness in improving hospital quality (Edmond & Bearman, 2007). Other research analyzes transparency mechanisms, such as how the publication of prescription data affects prescribing behaviors (Du, Wang, et al., 2015). Two public reporting programs were compared for their differing impacts on clinical and perceived quality (Han & Park, 2021). Research also examined how public reporting and market competition influence hospitals' technical efficiency (Han & Lee, 2021) and explored the relationship between patient experience and financial incentives to understand how public policy and healthcare practices can better promote patient-centered outcomes and choice (Schlesinger et al., 2015). Specifically, studies have explored how public reporting mechanisms affect general practitioners' prescribing behaviors, focusing on factors such as information availability, GP attitudes, perceived value, and perceived risk (Du, Zhang, et al., 2015).

Public reporting -Related Factors

Quality Indicators

Scholars have examined how public reporting influences individual choices in healthcare, finding that quality indicators play a significant role. Patients often consider structured data, such as survival and infection rates, when making choices (Duncanson et al., 2024). Studies also explore the impact of public reporting on quality improvement, particularly regarding process indicators like reperfusion rates and mortality rates for AMI patients (Chae et al., 2022), as well as the importance of quality indicators in selecting hospitals and physicians (Ellis et al., 2020). Other studies focus on how public reporting systems have improved healthcare quality and reduced costs in Wisconsin and California (Toussaint et al., 2014), or whether mandatory public reporting assumptions effectively enhance hospital quality (Edmond & Bearman, 2007). Research shows that public reporting can support consumer choice and encourage providers to improve care quality (J. B. Christianson et al., 2010) and that it has had positive effects on post-acute care quality in U.S. nursing homes (Werner et al., 2009), cancer surgery wait times (Prang et al., 2021), appropriate antibiotic use in URTI patients (Yang et al., 2014), and flu vaccination rates among healthcare workers (Paris et al., 2013). Moreover, public reporting has contributed to quality improvement in cardiovascular specialty care (Dehmer et al., 2014). Emerging strategies to improve vaccination rates as a quality and safety indicator for healthcare workers are also being explored (Johnson & Talbot, 2011), alongside home

healthcare quality improvement efforts (Jung K et al., 2010).

Performance Indicators

Studies have assessed whether performance indicators in long-term care facilities show improvement following public reporting and compared these trends with non-reported indicators (Poldrugovac et al., 2022). Research has also examined the information valued most by referring physicians in hospital quality reports, such as hospital performance and patient experience data, to better meet physicians' needs (Geraedts et al., 2018). Additionally, studies have shown that public reporting of healthcare performance data reveals disparities across insurance plans, encouraging improvement (Snowden AM et al., 2012). Comparative studies on evaluating provider performance (Paddock, 2014) and the impact of public reporting and pay-for-performance incentives on quality improvement further emphasize how reputational and financial incentives drive providers to elevate care standards. The effectiveness of public reporting data in understanding and enhancing rural Australian hospitals' operational performance has also been evaluated (Lloyd et al., 2021). Other studies investigate how public reporting influences physicians at various performance levels (Tang et al., 2017) and whether public reporting can improve German hospitals' quality without financial incentives (Bayindir & Schreyögg, 2023). Research on staff perceptions of publicly reported performance data highlights its role in internal decision-making and actions (Hafner JM et al., 2011).

Data Usage

Research has explored the feasibility of using state-level electronic health record (EHR) data to measure and reduce health disparities (Smith et al., 2023). Studies also assess consistency in hospital recommendations in Germany's report card system and the causes of ranking inconsistencies (Emmert et al., 2017). Evaluations of claims-based data's compliance with regulatory standards and its applications in hospital quality reporting and pay-for-performance programs have been conducted (Crews et al., 2016). To ensure accuracy and consistency in publicly reported infection data, standardized definitions and monitoring methods are recommended (Hebden, 2012). Infection Composite Scores (ICS) have been developed to help hospitals assess infection control performance over time, identify underperforming areas, and implement improvements. The use of data from the National Survey of Substance Abuse Treatment Services (N-SSATS) has been studied for its ability

to capture additional information relevant to accessibility and service coordination (Vaughn et al., 2020).

System Usage

The role of the Public Health Automated Case Event Reporting (PACER) system and FHIR interoperability standards in improving public health reporting, especially during rapid infectious disease outbreaks, has been examined (Mishra et al., 2020). Studies have developed systems to track critical health outcome indicators in head and neck cancer treatment to enhance clinical practice and public reporting (Walters et al., 2014). When analyzing the rating distribution on the public reporting website Jameda, researchers examined how patient and physician characteristics influence overall scores (Emmert & Meier, 2013).

Results of Robot Use in Health Care

Overview

We classified the antecedent based on the result. This led to the formation of three theme, individual related factors, organization related factors and public reporting related factors. For individual-related factors, we included evaluation tools and health literacy, while organization-related factors involved organizational attitudes, regional disparities, and adverse impacts. Public reporting-related factors focused on quality indicators and standardized frameworks. Ultimately, we categorized the studies into seven themes discussed in the following section.

individual related consequence

Evaluation Tools

Studies indicate a correlation between hospital evaluations and traditional quality metrics. Highly rated hospitals on social media often exhibit better quality metrics, though this correlation is not absolute (Lagu & Greaves, 2015). Low consistency in rankings among various report cards has also been observed (Emmert et al., 2017). Research suggests that physician rating websites significantly influence patients' physician selection, and this impact is likely to grow as public awareness increases (Emmert et al., 2013). The star rating system notably affects nursing homes' market shares, with consumers more likely to choose highly rated facilities (Werner et al., 2016). Similarly, public reporting influences consumer choices, with higher-rated home care facilities being more frequently

chosen after the introduction of public reporting (Jung et al., 2016). Ratings for female doctors tend to be higher than for their male counterparts, and older patients and those with private health insurance generally provide higher ratings (Emmert & Meier, 2013).

Health Literacy

Future public reporting developers should focus on user experience by creating reports tailored to different user groups (Shaller et al., 2014). Specific measures can increase minority awareness of public reporting (Yonek et al., 2014). Perceptions of public reporting are influenced by factors such as educational level, health status, and trust (J. Christianson et al., 2014). Patients with advanced dementia tend to perform poorly on specific quality indicators, particularly those related to functional decline and weight loss (Konetzka et al., 2015). MHCP patients generally score lower on quality indicators, with significant gaps persisting despite some improvements (Snowden AM et al., 2012). Results also indicate that users with no prior exposure to hospital report cards show a stronger intention to use them (Emmert & Wiener, 2017).

Organization related consequence

Attitudes

The availability of information, along with perceived value and attitudes of general practitioners, significantly impacts behavioral intentions, while perceived risk does not (Du, Zhang, et al., 2015). Acute care hospitals in Flanders have made substantial investments in quality improvement policies, with most hospitals completing initial certifications, publicly reporting quality indicators, and participating in pay-for-performance programs. Despite this, stakeholder opinions on these policies vary (Van Wilder et al., 2021). Public reporting of HAI data varies significantly across European countries; experts in countries with established reporting systems generally support it, while experts in countries without such systems express reservations. Experts also suggest that reporting structure and process indicators is more beneficial for hospital quality improvement than direct infection rates, due to concerns about data misinterpretation and biases (Martin M et al., 2013). Providers express concerns about how transparency may affect reputation and market share (Lihn et al., 2015). Many physicians view quality data on online resources with skepticism, doubting its accuracy. Only 2%

consider Medicare's public reporting site highly accurate, and over half believe the data is not helpful for patients choosing doctors. They prefer foundational information, such as board certifications and clinical interests, over performance-based quality metrics or patient reviews (Lagu et al., 2019). Studies suggest that hospital report cards should emphasize physician and patient experiences to better support consultations. This implies that presenting hospital quality data requires adjustments to maximize its utility in clinical decision-making (Geraedts et al., 2018). The availability of information and the perceived value and attitudes of general practitioners significantly influence their behavioral intentions, whereas perceived risk does not exert a significant effect (Du, Zhang, et al., 2015).

Regional Disparities in Improvement

Evidence suggests that public reporting has had a significant impact on performance improvement (Poldrugovac et al., 2022). Public reporting transparency policies can enhance social welfare by reducing inequalities and encouraging hospital investments in quality improvement (Saghafian & Hopp, 2020). Transparency efforts have also improved prescribing behaviors and appropriate drug use (Du, Wang, et al., 2015). The NPC-QIC initiative has achieved significant advancements in reducing infant mortality, improving nutrition management, and optimizing care processes (Lihn et al., 2015). In Wisconsin and California, public reporting systems have significantly improved healthcare service quality while reducing costs (Toussaint et al., 2014). However, while higher-performing hospitals have benefited from public reporting, lower-performing hospitals have not shown similar improvements (Bayindir & Schreyögg, 2023). Studies have also noted variation in improvements across institutions, with some metrics showing significant advancement while others have lagged (Jung K et al., 2010), performance data from public reporting (Lloyd et al., 2021). In one survey, 73% of PICU medical directors believed that benchmarking could improve patient outcomes (Natale et al., 2011). The impact of transparency is more pronounced for younger, wealthier, or urban patients compared to rural or older patients (Saghafian & Hopp, 2020). Given the complex effects of public reporting on hospital efficiency across different market environments, policymakers are advised to consider market dynamics to achieve optimal healthcare outcomes (Han & Lee, 2021). Studies also show that while some public reporting programs have improved clinical quality in specific healthcare domains, others primarily affect public perception rather than

actual quality outcomes (Han & Park, 2021). In specific cases, such as elective surgery for lung cancer, waiting times have slightly increased (Prang et al., 2021).

Adverse Effects

Public reporting can also lead to unintended consequences, sometimes termed “metric-driven harm” (Rambur et al., 2013). While patients generally support the public reporting of performance data, they also report increased anxiety, especially when they are unable to change their healthcare provider. Many patients prefer data to be presented positively, with explanations and educational resources to help them understand and make informed decisions. Additionally, patients emphasize the importance of trusting their healthcare team and often rely more on physician advice than on statistical data alone (Duncanson et al., 2024). While quality competition is credited with improving healthcare standards in Germany, quality measurement and public reporting are seen as among the most effective tools in this regard (Negele et al., 2023). Many studies recommend that regulators prioritize incentives for hospitals to improve quality, considering interventions that encourage hospitals to enhance their services (Saghafian & Hopp, 2020). However, despite the transparency and accountability that public reporting promotes, unintended negative outcomes may arise. For instance, some physicians may avoid high-risk patients to improve their reported performance metrics. Furthermore, doctors and hospitals may focus more on meeting specific “target rates” rather than providing individualized care based on clinical judgment and patient needs. While some studies suggest that public reporting has contributed to lower cardiac surgery mortality rates, others find evidence that some physicians may refuse high-risk cases to maintain favorable evaluations. Additionally, while transparency has generally improved healthcare quality, studies reveal its impact can be limited, and in certain cases, it may even inadvertently reduce care quality (Werner & Asch, 2005). Public reporting requirements for PCI have imposed a substantial financial burden on hospitals, with catheterization labs incurring annual costs between \$100,000 and \$200,000, with some spending even higher amounts (Lagu et al., 2019). Although incentive-based programs have sometimes threatened aspects of patient-centered care, combining these incentives with patient experience reports and outcomes has shown to enhance patient-valued health outcomes (Schlesinger et al., 2015).

Public reporting -Related Factors

Quality Indicators in Public Reporting

Public reporting has been shown to positively impact quality indicators and process measures, such as reperfusion rates and AMI mortality (Chae et al., 2022), and reduce antibiotic prescription rates (Tang et al., 2017)(Yang et al., 2014). Meanwhile, CLC quality improvements (Davila et al., 2024)(Pimentel et al., 2021), vaccination rates (Paris et al., 2013), quality of care metrics (Hafner JM et al., 2011)(Werner et al., 2009), infection control measures (Edmond & Bearman, 2007)(Fakih et al., 2016), and cardiovascular care quality (Dehmer et al., 2014) have all improved following public reporting implementation. However, public reporting systems may require adjustments to focus on factors patients find meaningful, such as hospital reputation and physician expertise, to increase their usage and impact (Ellis et al., 2020).

Standardized Framework

Public reporting's influence on quality improvement has been limited, especially during the late 1990s, as third-party payers, patients, and referring physicians made little use of publicly available data for clinical decisions (Rhoads et al., 2009). A unified data processing framework could address these issues by providing an effective approach to measuring and monitoring health disparities (Smith et al., 2023). Collaborative efforts between public health agencies and healthcare systems have enabled faster case information collection, reduced human errors, and ensured data privacy and accuracy (Mishra et al., 2020). Future systems should support automated, structured clinical data entry to effectively track and report treatment outcomes, thereby enhancing patient care and supporting a value-based healthcare system (Walters et al., 2014). Significant discrepancies in monitoring practices across institutions suggest the need for standardized definitions and automated systems to improve data accuracy and consistency (Hebden, 2012). Standardized and validated reporting methods are also essential for improving report credibility and applicability (Johnson & Talbot, 2011). Integration of patient experience data and real-time information is increasingly considered essential for the future of public reporting (J. B. Christianson et al., 2010). A new survey framework has been proposed to guide data collection and improve services in the future (Vaughn et al., 2020).

The format of data presentation in public reports, such as using charts or text and including confidence intervals, does not significantly impact consumer understanding (Mazor KM et al., 2009). However, providing detailed explanations within public reporting data enhances individuals' ability to comprehend and make informed choices (Duncanson et al., 2024). Supporting consumers in better understanding performance data is essential. Seven elements of information presentation can improve user comprehension, such as symbol-annotated bar charts, clear labels indicating high and low values relative to hospital performance, and rankings based on performance (Sander et al., 2015).